

gat-typst Package Manual

Version 0.1.0

A comprehensive Typst package for academic writing, homework assignments, and cheatsheets

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1 Introduction

The `gat-typst` package is a comprehensive template system for academic writing in Typst, with a focus on mathematics-heavy homework assignments, cheatsheets, and technical documents. It provides opinionated defaults, extensive math utilities, and a curated collection of the best packages from the Typst ecosystem.

This is a growing library for Gatlen Culp's personal usage and is unstable.

1.1 Key Features

- **Templates:** Ready-to-use templates for homework and cheatsheets
- **Math Utilities:** Enhanced math notation with semantic symbols and custom functions
- **Optional Styling:** Professional defaults for code, tables, equations, and math shorthands (opt-in via style)
- **Convenience Functions:** Colored boxes, custom math operators, and more
- **Integrated Packages:** Core packages for math, code, and formatting

1.2 Installation

To use this package locally, ensure it's installed in your local Typst packages directory:

```
1 # Unix-like systems
2 ~/.local/share/typst/packages/local/gat-typst/0.1.0/
3
4 # Windows
5 %APPDATA%/typst/packages/local/gat-typst/0.1.0/
```

Then import it in your Typst documents:

```
1 #import "@local/gat-typst:0.1.0": *
```

1.3 Quick Start

Here's a minimal homework document to get started:

```
1 #import "@local/gat-typst:0.1.0": *
2
3 #show: homework.with(
4   title: "Homework 1",
5   course: "CS 101",
6   university: "Your University",
7   author: "Your Name",
8   email: "you@example.com",
9   date: datetime.today(),
10 )
11
12 // Optional: Enable enhanced styling (math shorthands, code styling, etc.)
13 #show: style.with()
14
15 #question(status: "done")[
16   Prove that  $P = N P$ .
```

```
17 ]
18
19 #solution[
20     Left as an exercise to the reader.
21
22     Using our custom math utilities:
23      $E(X^2) \geq (E(X))^2$  by Jensen's inequality.
24 ]
```

2 Important Behavior Changes

When using the `style` function, several default behaviors are modified for improved mathematical typesetting and consistency. **These changes are opt-in** - you must explicitly use `#show: style.with()` to enable them.

2.1 Math Shorthands

The `style` function enables extensive math operator shorthands via the `quick-maths` package.

2.1.1 Complete Shorthand List

Operators:

1	<code>\$a ** b\$</code>	// Small asterisk	Typst
2	<code>\$a * b\$</code>	// Centered dot ⚠	
3	<code>\$a +- b\$</code>	// Plus-minus	
4	<code>\$A - B\$</code>	// Turnstile	
5	<code>\$a o+ b\$</code>	// Circled plus	
6	<code>\$a o. b\$</code>	// Circled dot	
7	<code>\$a @ b\$</code>	// Also circled dot	

$a * b$ $a \cdot b$ $a \pm b$

$A \vdash B$ $ao + b$

$ao.b$ $a@b$

Logic:

1	<code>\$a .. b\$</code>	// Spacing (quad)	Typst
2	<code>\$A :. B\$</code>	// Therefore	
3	<code>\$A :' B\$</code>	// Because	

$a \quad b$

$A \therefore B$ $A \because B$

Relations:

1	<code>\$a -<= b\$</code>	// Precedes or equals	Typst
2	<code>\$a -< b\$</code>	// Precedes	
3	<code>\$a >-= b\$</code>	// Succeeds or equals	
4	<code>\$a >- b\$</code>	// Succeeds	
5	<code>\$a -= b\$</code>	// Equivalent	
6	<code>\$a ~= b\$</code>	// Asymptotically equal	
7	<code>\$a ~~ b\$</code>	// Approximately	

$a \preceq b$ $a \prec b$

$a \succeq b$ $a \succ b$

$a \equiv b$ $a \cong b$ $a \approx b$

Brackets:

1	<code>\$<< a, b >>\$</code>	// Angle brackets	Typst
2	<code>\$a == b\$</code>	// Equals (unchanged)	

$\ll a, b \gg$

$a = b$

2.1.2 Warning About * Shorthand

The `* → dot.op` replacement is the most impactful change:

1	// With shorthands (default):	Typst
2	<code>\$a * b\$</code>	// Renders: $a \cdot b$
3	<code>\$a ** b\$</code>	// Renders: $a * b$ (small asterisk)
4		

```
5 // To use literal asterisk:
6 $a ast b$           // Renders: a * b
```

$a \cdot b$ (renders as $a \cdot b$)

$a * b$ (small asterisk)

$a \cdot b$ (literal asterisk)

This follows conventions in linear algebra and physics but may surprise users expecting traditional asterisk multiplication.

2.1.3 Disabling Shorthands

Shorthands are disabled by default. To enable them, use:

```
1 #show: style.with()
```

If you want to use style without shorthands, you can disable them:

```
1 #show: style.with(enable-shorthands: false)
```

Note: Currently this parameter exists but shorthands are applied regardless when style is used. This will be fixed in a future version.

2.1.4 Examples

```
1 // Statistics
2 $Ex(X) .. Var(X) = Ex(X^2) - (Ex(X))^2$
3
4 // Linear algebra
5 $A * B \neq B * A .. "in general"$
6
7 // Logic
8 $x > 0 \therefore x^2 > 0$
9
10 // Approximation
11 $e \approx 2.718$
12
13 // Relations
14 $x \preceq y \text{ "iff" } f(x) \leq f(y)$
```

$$\mathbb{E}[X] \quad \text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

$$A \cdot B \neq B \cdot A \quad \text{in general}$$

$$x > 0 \therefore x^2 > 0$$

$$e \approx 2.718$$

$$x \preceq y \text{ iff } f(x) \leq f(y)$$

2.2 Superscript Transformations

Three special superscript transformations are enabled for common mathematical notation.

2.2.1 Transpose (T)

```
1 $A^T = "transpose of" A$
2 $x^T y = \text{iprod}(x, y)$
3 $(A B)^T = B^T A^T$
```

$$A^T = \text{transpose of } A$$

$$x^T y = \langle x, y \rangle$$

$$(AB)^T = B^T A^T$$

Renders with proper transpose symbol (T) instead of superscript T.

2.2.2 Dagger ($^+$)

```
1 $A^+ = "Moore-Penrose
```

```
pseudoinverse"$
```

```
2 $H^+ = "Hermitian transpose"$
```

$A^\dagger$ = Moore-Penrose pseudoinverse

$H^\dagger$ = Hermitian transpose

Renders as dagger symbol ($\dagger$).

2.2.3 Star (^*)

```
1 $V^* = "optimal value function"$  
2 $pi^* = "optimal policy"$  
3 $x^* = arg min_x f(x)$
```

V^* = optimal value function

π^* = optimal policy

$x^* = \arg \min_x f(x)$

Renders with star symbol ($\star$) instead of superscript asterisk.

This distinguishes from the asterisk operator and provides cleaner mathematical notation.

2.2.4 Interaction with Shorthands

The superscript transformations work correctly even when $\star$ shorthand is enabled:

```
1 $A^T * B^*$ // Transpose A  
times optimal B
```

$A^T \cdot B^\star$

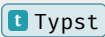
3 Utilities

The package provides extensive utilities for math, boxes, and display formatting.

3.1 Math Utilities

3.1.1 Statistics Functions

Colored, bracketed notation for statistical operators:

1	<code>\$Ex(X)\$</code>	// Expected value	
2	<code>\$Var(X)\$</code>	// Variance	
3	<code>\$Cov(X, Y)\$</code>	// Covariance	
4	<code>\$Corr(X, Y)\$</code>	// Correlation	
5	<code>\$Pr(X > 0)\$</code>	// Probability	

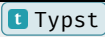
$\mathbb{E}[X]$ $\text{Var}[X]$ $\text{Cov}[X]$
 $\text{Corr}[X]$ $\text{Pr}[X > 0]$

These render with blue coloring and square brackets. Example:

$\mathbb{E}[X^2] \geq (\mathbb{E}[X])^2$ by Jensen's inequality.

$\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y]$ when X and Y are independent.

3.1.2 Reasoning Operators

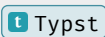
1	<code>\$x st x > 0\$</code>	// "such that"	
2	<code>\$If x > 0 Then y = 1 Else y = 0\$</code>		

$x \text{ s.t. } x > 0$

if $x > 0$ then $y = 1$ else $y = 0$

Example: $\max\{x \text{ s.t. } x \in \mathbb{R}\}$

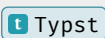
3.1.3 Machine Learning / RL

1	<code>\$KL(p, q)\$</code>	// KL divergence	
2	<code>\$refPol\$</code>	// Reference policy	

$D_{\text{KL}}(p \parallel q)$ $\bar{\pi}$

Example: Minimize $D_{\text{KL}}(\pi \parallel \bar{\pi})$ to stay close to reference policy.

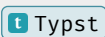
3.1.4 Miscellaneous

1	<code>\$SE\$</code>	// Standard error	
2	<code>\$ddot\$</code>	// Double dot	
3	<code>\$alias\$</code>	// Alias arrow	
4	<code>\$clamp(x)\$</code>	// Clamp function	
5	<code>\$solve(x)\$</code>	// Solve function	

SE $\overset{\text{alias}}{\leftarrow}$
 $\text{clamp}\{x\}$ $\text{solve}\{x\}$

3.1.5 Custom Math Function Factory

Create your own math functions with custom delimiters:

1	<code>#let myFunc = math-</code>	
	<code>func("myFunc", delim: "[")</code>	
2	<code>\$myFunc(x, y)\$</code>	
3		
4	<code>// Or with custom left/right delimiters</code>	

```
#let norm = math-func("norm", ldelim:  
5 math.bar.v.double, rdelim:  
  math.bar.v.double)  
6 $norm(x)$
```

`myFunc[x]`

`norm||x||`

3.2 Box Utilities

3.2.1 gatbox()

Customizable colored boxes with automatic color cycling.

3.2.1.1 Basic Usage

```
1 #gatbox[ t Typst
2   This is a basic box with default black
   color.
3 ]
4
5 #gatbox(color: blue)[
6   This is a blue box.
7 ]
```

This is a basic box with default black color.

This is a blue box.

3.2.1.2 Color Cycling

Automatically cycle through predefined colors:

```
1 #gatbox(color-cycle: true)[First t Typst
   box - blue]
2 #gatbox(color-cycle: true)[Second box -
   red]
3 #gatbox(color-cycle: true)[Third box -
   green]
4 #gatbox(color-cycle: true)[Fourth box -
   orange]
5 #gatbox(color-cycle: true)[Fifth box -
   purple]
```

First box - blue

Second box - red

Third box - green

Fourth box - orange

Fifth box - purple

3.2.1.3 Parameters

- `color` (color, default: black): Box color
- `color-cycle` (boolean, default: false): Auto-cycle through colors
- `title` (content): Optional title
- All `showybox` parameters supported

3.2.1.4 Examples

```
1 #gatbox(title: "Important", t Typst
   color: red)[
2   This is a critical point!
3 ]
4
5 #gatbox(color-cycle: true, title:
   "Theorem 1")
```

```
6 All horses are the same color.
7 ]
```

Important

This is a critical point!

Theorem 1

All horses are the same color.

3.2.2 showybox

The base box system (re-exported from the `showybox` package) with `breakable: true` by default.

```
1 #showybox(t Typst  
2   title: "Note",  
3   frame: (  
4     border-color: blue,  
5     title-color: blue.lighten(80%),  
6   ),  
7 ) [  
8   Content here can break across pages.  
9 ]
```

Note

Content here can break across pages.

3.3 Display Utilities

3.3.1 Show Rules

The package includes custom show rules that can be used independently:

```
1 #show: super-ast-as-start Typst  
2  
3 $V^* = V(x^*)$ // Asterisk renders as  
   star symbol
```

$$V^* = V(x^*)$$

4 Style Function

The `style` function provides enhanced styling for code, tables, and math notation. This is **opt-in** and must be explicitly enabled with `#show: style.with()`.

4.1 Code Styling

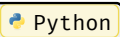
When using `style`, all code blocks are automatically styled:

4.1.1 Automatic Changes

1. **Font size:** Reduced to 8pt for compactness
2. **Zebra striping:** Alternating row backgrounds (`gray.lighten(80%)`)
3. **Border:** 1pt black stroke around code blocks
4. **Syntax highlighting:** Language-specific coloring

4.1.2 Example

```
1 def quicksort(arr):
2     if len(arr) ≤ 1:
3         return arr
4     pivot = arr[len(arr) // 2]
5     left = [x for x in arr if x < pivot]
6     middle = [x for x in arr if x == pivot]
7     right = [x for x in arr if x > pivot]
8     return quicksort(left) + middle + quicksort(right)
```



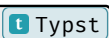
No manual configuration needed - just use standard Typst raw blocks with language tags.

4.2 Table Styling

When using `style`, the `booktabs` package is automatically applied, giving all tables professional styling with:

- Proper spacing between rows
- Professional horizontal rules (top, mid, bottom)
- No vertical lines (following `booktabs` philosophy)
- Better visual hierarchy

```
1 #table(
2     columns: 3,
3     [*Algorithm*], [*Time*], [*Space*],
4     [Quicksort], [$O(n \log n)$], [$O(\log n)$],
5     [Mergesort], [$O(n \log n)$], [$O(n)$],
6     [Heapsort], [$O(n \log n)$], [$O(1)$],
7 )
```



Algorithm	Time	Space
Quicksort	$O(n \log n)$	$O(\log n)$
Mergesort	$O(n \log n)$	$O(n)$
Heapsort	$O(n \log n)$	$O(1)$

No special syntax required - standard `#table()` gets enhanced styling.

5 Templates

The package provides two main templates: `homework` and `cheatsheet`. Both templates can be enhanced with the optional `style` function for additional styling features.

5.1 Style Function

The `style` function provides enhanced styling for code, tables, equations, and math notation. It is **optional** and must be explicitly enabled.

5.1.1 Usage

```
1 #show: style.with()
```

 Typst

5.1.2 Parameters

- `enable-shorthands` (boolean, default: `true`): Enable math operator shorthands

5.1.3 Features Provided

The `style` function automatically configures:

1. **FontAwesome Icons:** Access to icon library via `fa-icon()` function
2. **Code Styling:** Syntax highlighting with Codly, 8pt font size, zebra striping
3. **Table Styling:** Professional booktabs-style tables by default
4. **Math Configuration:**
 - Equation numbering with `equate` package
 - Breakable equations
 - Sub-numbering disabled
5. **Show Rules:**
 - T renders as transpose
 - $^+$ renders as dagger ($\dagger$)
 - * renders as star ($\star$) instead of asterisk

5.1.4 Example

```
1 #import "@preview/gat-typst:0.1.0": *
2
3 #show: style.with()
4
5 // Use FontAwesome icons
6 #fa-icon("heart") Math is beautiful
7
8 // Code with automatic styling
```

 Typst

```
python def hello(): print("Hello, world!")
```

```
1
2 // Professional tables
3 #table(
4   columns: 3,
5   [*Name*], [*Age*], [*City*],
6   [Alice], [25], [Boston],
```

```
7     [Bob], [30], [Seattle],
8 )
9
10 // Math with special superscripts
11 $ A^T dot B^+ = C^* $
```

♥ Math is beautiful

```
1 def hello():
2     print("Hello, world!")
```

 Python

Name	Age	City
Alice	25	Boston
Bob	30	Seattle

$$A^T \cdot B^\dagger = C^\star$$

5.2 Homework Template

The homework template creates professional homework assignments with custom components for questions, solutions, and more.

5.2.1 Usage

```
1  #show: homework.with(t Typst
2    title: "Homework 1",
3    course: "Course Name",
4    university: "University Name",
5    author: "Your Name",
6    email: "your@email.com",
7    collaborators: [List of collaborators],
8    resources: [Resources used],
9    date: datetime.today(),
10   cols: 1, // or 2 for two-column layout
11   show-header: true,
12   paper-size: "us-letter",
13 )
```

5.2.2 Parameters

- title (string): Homework title (e.g., “HW01”, “Problem Set 3”)
- course (content): Course name or code
- university (content): University name
- author (content): Your name
- email (content): Your email address
- collaborators (content, default: []): List of collaborators
- resources (content, default: []): Resources used
- date (datetime, default: datetime.today()): Assignment date
- cols (integer, default: 1): Number of columns (1 or 2)
- show-header (boolean, default: true): Show header with metadata
- paper-size (string, default: "us-letter"): Paper size

5.2.3 Components

5.2.3.1 question()

Creates a colored box for problem statements with optional status tracking.

```
1  #question(status: "done")[t Typst
2    What is the derivative of  $x^2$ ?
3  ]
4
5  #question(status: "partial")[
6    This is partially complete.
7  ]
8
9  #question(status: "todo")[
10   Not started yet.
```



```

11 ]
12
13 #question()[
14   No status indicator.
15 ]

```

Status colors:

- "done": Green
- "partial": Yellow
- "todo": Red
- "none" or omitted: Black

5.2.3.2 solution()

Wrapper for solution content with optional column layout.

```

1 #solution[
2   The derivative is  $2x$ .
3 ]
4
5 #solution(cols: 2)[
6   This solution spans two columns.
7 ]

```

 Typst

5.2.3.3 boxed-solution

Inline box for short solutions.

```

1 The answer is #boxed-solution[$x = 5$].

```

 Typst

5.2.3.4 sub-solution

Nested solution box for multi-part problems.

```

1 #sub-solution[
2   For part (a), we first note that ...
3 ]

```

 Typst

5.2.3.5 notes

Purple box for additional notes or remarks.

```

1 #notes[
2   Remember to check edge cases!
3 ]

```

 Typst

5.2.4 Complete Example

```

1 #import "@local/gat-typst:0.1.0": *
2
3 #show: homework.with(
4   title: "HW02",
5   course: "Linear Algebra",

```

 Typst

```

6   university: "MIT",
7   author: "Student Name",
8   email: "student@mit.edu",
9   collaborators: [Alice, Bob],
10  resources: [Textbook Chapter 3],
11  date: datetime(year: 2025, month: 1, day: 15),
12  cols: 1,
13 )
14
15 = Matrix Operations
16
17 #question(status: "done")[
18   Let  $A = \text{mat}(1, 2; 3, 4)$ . Compute  $A^T A$ .
19 ]
20
21 #solution[
22   We compute:
23   $
24    $A^T A = \text{mat}(1, 3; 2, 4) \text{mat}(1, 2; 3, 4)$ 
25    $= \text{mat}(10, 14; 14, 20)$ 
26   $
27
28   #sub-solution[
29     Note that  $A^T A$  is always symmetric.
30   ]
31 ]

```

5.3 Cheatsheet Template

The cheatsheet template creates compact, three-column cheatsheets perfect for exams.

5.3.1 Usage

```
1 #show: cheatsheet.with(t Typst
2   title: "Midterm",
3   course: "Course Name",
4   university: "University Name",
5   author: "Your Name",
6   email: "your@email.com",
7   date: datetime.today(),
8 )
```

5.3.2 Parameters

Similar to homework but without cols, show-header, and paper-size (fixed to 3-column layout on US Letter with 0.5cm margins).

5.3.3 Features

- **3-column layout** with 0.3cm gutter
- **Compact spacing**: 6pt base font, 7pt level-1 headings, 6pt level-2/3
- **No page numbers**: Maximizes space
- **Centered headings**: Better visual organization

5.3.4 Example

```
1 #import "@local/gat-typst:0.1.0": *t Typst
2
3 #show: cheatsheet.with(
4   title: "Final Exam Cheatsheet",
5   course: "Calculus I",
6   author: "Your Name",
7   date: datetime(year: 2025, month: 5, day: 10),
8 )
9
10 = Derivatives
11
12 == Power Rule
13 $ dv(, x) x^n = n x^(n-1) $
14
15 == Product Rule
16 $ dv(, x) (f g) = f' g + f g' $
17
18 == Chain Rule
19 $ dv(, x) f(g(x)) = f'(g(x)) g'(x) $
20
21 = Integrals
22
23 == Power Rule
```

$$24 \quad \int x^n dx = \frac{x^{n+1}}{n+1} + C$$

6 Dependencies & Imports

The package integrates carefully selected external packages. The `style` function automatically imports core packages for math, code, and formatting.

6.1 Overview

Active imports from `src/style/style.typ` (when using `#show: style.with()`):

```
1 // == Math ==
2 #import "@preview/equate:0.3.2": equate
3 #import "@preview/physica:0.9.6": *
4 #import "@preview/quick-maths:0.2.1": shorthands
5
6 // == Code ==
7 #import "@preview/codly:1.3.0": *
8 #import "@preview/codly-languages:0.1.1": codly-languages
9
10 // == Display & Formatting ==
11 #import "@preview/fontawesome:0.6.0": *
12 #import "@preview/booktabs:0.0.4": *
13
14 // == Utilities ==
15 // Available through gat-typst but not auto-imported by style:
16 // #import "@preview/showybox:2.0.4": showybox
17 // #import "@preview/hydra:0.6.2": hydra
18
19 // == Diagramming == (not included in style, must import separately)
20 // #import "@preview/cetz:0.4.2": canvas, draw
21 // #import "@preview/cetz-plot:0.1.2": plot
22 // #import "@preview/fletcher:0.5.8" as fletcher: diagram, edge, node
23 // #import "@preview/lilaq:0.5.0" as lq
24
25 // == Additional Math == (not included in style)
26 // #import "@preview/matset:0.1.0"
27 // #import "@preview/mannot:0.3.0": *
28
29 // == Additional Code == (not included in style)
30 // #import "@preview/callisto:0.2.4"
31 // #import "@preview/lovelace:0.3.0": pseudocode-list
32 // #import "@preview/cmarker:0.1.6"
33
34 // == Additional Display == (not included in style)
35 // #import "@preview/frame-it:1.2.0": *
36
37 // == Compatibility == (not included in style)
38 // #import "@preview/mitex:0.2.5": mitex, mitex-convert, mitext
39
40 // == Color == (not included in style)
41 // #import "@preview/splash:0.5.0": xcolor
```

```
42 // #import "@preview/typpuccino:0.1.0": frappe, latte, macchiato, mocha
```

6.2 Diagramming

Note: Diagramming packages are not included in **style**. Import them explicitly if needed.

6.2.1 CeTZ (v0.4.2)

Comprehensive drawing library for creating custom diagrams.

```
1 #import "@preview/cetz:0.4.2": canvas, draw
2
3 #canvas({
4   import draw: *
5   circle((0, 0), radius: 1)
6   line((0, 0), (1, 1))
7   content((0, 0), $x$)
8 })
```

Use for: Custom diagrams, network architectures, geometric figures.

6.2.2 CeTZ Plot (v0.1.2)

Plotting extension for CeTZ.

```
1 #import "@preview/cetz-plot:0.1.2": plot
2
3 #canvas({
4   plot.plot(
5     size: (8, 6),
6     x-tick-step: 1,
7     y-tick-step: 1,
8     {
9       plot.add(x => x * x, domain: (-2, 2))
10    }
11  )
12 })
```

Use for: Function plots, data visualization.

6.2.3 Fletcher (v0.5.8)

Arrow diagrams and state machines.

```
1 #diagram(
2   node((0, 0), $A$, radius: 2em),
3   node((1, 0), $B$, radius: 2em),
4   edge((0, 0), (1, 0), "→", $f$),
5 )
```

Use for: State machines, flowcharts, commutative diagrams.

6.2.4 Lilaq (v0.5.0)

Modern data visualization library.

```
1 #lq.diagram(  
2   title: [Data Plot],  
3   xlabel: $x$,  
4   ylabel: $y$,  
5   lq.plot((0, 1, 2, 3), (1, 4, 2, 3), mark: "o"),  
6 )
```

Use for: Scientific plots, cleaner syntax than CeTZ Plot.

6.3 Math

Note: **Equate**, **Physica**, and **Quick-maths** are automatically imported by **style**. Other math packages must be imported explicitly.

6.3.1 Matset (v0.1.0)

Not included in **style** - import explicitly if needed.

Set notation support.

Use for: Mathematical set operations and notation.

6.3.2 Equate (v0.3.2)

Included in **style** - automatically available.

Advanced equation numbering with breakable equations.

```
1 $  
2   f(x) = x^2  
3 $ <eq:quadratic>  
4  
5 Reference: @eq:quadratic
```

$$f(x) = x^2$$

Use for: Equation numbering and cross-referencing.

6.3.3 Physica (v0.9.6)

Included in **style** - automatically available.

Comprehensive physics notation package.

Key functions:

```
1 $vb(x)$           // Vector  
   bold  
2 $vu(u)$           // Unit vector  
3 $dd(x)$           // Differential  
4 $dv(f, x)$        // Derivative  
5 $pdv(f, x, y)$    // Partial  
6 $grad f$, $div vb(F)$, $curl vb(F)$  
7 $iprod(u, v)$     // Inner product  
8 $norm(x)$         // Norm
```

```
9 $abs(x)$          // Absolute value  
10 $Set(x, x > 0)$  // Set builder  
11 $dmat(1, 2, 3)$   // Diagonal matrix  
12 $evaluated(f)_a^b$ // Evaluation bar
```

$x \quad \hat{u} \quad dx$

$\frac{df}{dx} \quad \frac{\partial^2 f}{\partial x \partial y}$

$\nabla f \quad \nabla \cdot F \quad \nabla \times F$

$\langle u, v \rangle \quad \|x\| \quad |x|$

$\{x \mid x > 0\}$

$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

$f|_a^b$

Example:

```
1 $ t Typst
2   dv(vb(x), t) = vb(v), quad
3   pdv(f, x, y, [2, 1]) = 0, quad
4   iprod(vb(u), vb(v)) = norm(vb(u))
   norm(vb(v)) cos theta
5 $
```

$\frac{dx}{dt} = v, \quad \frac{\partial^3 f}{\partial x^2 \partial y} = 0, \quad \langle u, v \rangle = \|u\| \|v\| \cos \theta$

Use for: Almost all physics and advanced math notation.

6.3.4 Quick-maths (v0.2.1)

Included in **style** - automatically available.

Math shorthand system (powers the operator replacements).

See “Important Behavior Changes” section for details.

6.3.5 Mannot (v0.3.0)

Not included in **style** - import explicitly if needed.

Math annotations and labels.

TODO: FIX

```
1 $ t Typst
2   p_i = exp(-beta E_i) / sum_j exp(-beta
   E_j)
3   #annot(<beta>, pos: top)[Inverse
   temperature]
4 $
```

Use for: Annotating complex equations, teaching materials.

6.4 Code

Note: Codly and Codly Languages are automatically imported by **style**. Other code packages must be imported explicitly.

6.4.1 Callisto (v0.2.4)

Not included in **style** - import explicitly if needed.

Jupyter notebook rendering.

```
1 #let (render, Cell, In, Out) = callisto.config(Typst
2   nb: json("notebook.ipynb"),
3 )
4
5 #In("cell-tag") // Show input cell
6 #Out("cell-tag") // Show output cell
7 #Cell("cell-tag") // Show both
```

Use for: Including Jupyter notebook code/output in documents.

6.4.2 Lovelace (v0.3.0)

Not included in **style** - import explicitly if needed.

Pseudocode formatting.

```
1 #pseudocode-list[Typst
2   + *Input:* Array  $A$  of length  $n$ 
3   + *Output:* Sorted array
4   + *for*  $i = 1$  *to*  $n$  *do*
5     + Process element  $A[i]$ 
6 ]
```

Use for: Algorithm descriptions.

6.4.3 Codly (v1.3.0) + Codly Languages (v0.1.1)

Included in **style** - automatically available.

Modern syntax highlighting for code blocks.

When using style, automatically configured with:

- 8pt font size
- 1pt black stroke
- Zebra striping (gray.lighten(80%))

```
1 def fibonacci(n):Python
2   if n ≤ 1:
3     return n
4   return fibonacci(n-1) + fibonacci(n-2)
```

Supports: Python, Rust, JavaScript, TypeScript, Java, C++, and 50+ more languages.

Use for: All code snippets. Automatically applied to raw blocks.

6.4.4 Cmarker (v0.1.6)

Not included in **style** - import explicitly if needed.

CommonMark (Markdown) support.

Use for: Converting Markdown to Typst.

6.5 Display & Formatting

Note: Font Awesome and Booktabs are automatically imported by **style**. Other display packages must be imported explicitly.

6.5.1 Frame-it (v1.2.0)

Not included in **style** - import explicitly if needed.

Custom frame styles for content boxes.

```
1 #let (example, definition) = frames(  
2   example: ("Example", blue),  
3   definition: ("Definition", green),  
4 )  
5  
6 #example[Title][Tag][  
7   Body content here.  
8 ]
```

 Typst

Use for: Theorem environments, examples, definitions.

6.5.2 Font Awesome (v0.6.0)

Included in **style** - automatically available.

Font Awesome icon support (version 6).

```
1 #fa-icon("heart")  
2 #fa-icon("calculator")  
3 #fa-icon("robot")
```

 Typst

Use for: Visual indicators, status icons. Note: Requires Font Awesome fonts installed.

6.5.3 Showybox (v2.0.4)

Available through gat-typst utilities (base for **gatbox**).

Styled, breakable boxes.

```
1 #showybox(  
2   title: "Title",  
3   frame: (border-color: blue),  
4   breakable: true,  
5 )[  
6   Content  
7 ]
```

 Typst

Use for: Custom boxes, callouts, highlighted content.

6.5.4 Hydra (v0.6.2)

Available through gat-typst but not included in **style**.

Context-aware page headers showing current section.

```
1 #import "@preview/hydra:0.6.2": hydra
```

 Typst

```

2
3 #set page(header: context [
4   #text(fill: gray)[**hydra(2)**]
5 ])

```

Use for: Displaying current section in headers. Used by the homework template.

6.5.5 Booktabs (v0.0.4)

Included in **style** - automatically available.

Publication-quality table styling.

When using **style**, automatically applied to all tables. Creates professional spacing and rules.

Use for: All tables. No manual configuration needed when using **style**.

6.6 Compatibility

Note: Compatibility packages are not included in **style. Import them explicitly if needed.**

6.6.1 MiTeX (v0.2.5)

Not included in **style** - import explicitly if needed.

LaTeX compatibility layer for including LaTeX code.

```

1 #mitex(`
2   \begin{bmatrix}
3     1 & 2 \\
4     3 & 4
5   \end{bmatrix}
6 `)

```

Typst

Use for: Converting existing LaTeX, complex LaTeX-only constructs.

6.7 Color

Note: Color packages are not included in **style. Import them explicitly if needed.**

6.7.1 Splash (v0.5.0)

Not included in **style** - import explicitly if needed.

Extended color support including xcolor names.

```

1 #text(fill: xcolor("red!50!blue"))[Mixed color]

```

Typst

Use for: Advanced color specifications.

6.7.2 Typpuccino (v0.1.0)

Not included in **style** - import explicitly if needed.

Catppuccin color schemes (frappe, latte, macchiato, mocha).

```

1 #import "@preview/typpuccino:0.1.0": mocha

```

Typst

```
2  
3 #text(fill: mocha.red)[Catpuccin red]
```

Use for: Consistent color theming.

7 Complete Examples

7.1 Simple Homework Document

```
1  #import "@local/gat-typst:0.1.0": *
2
3  #show: homework.with(
4    title: "Problem Set 1",
5    course: "Calculus II",
6    university: "Your University",
7    author: "Your Name",
8    email: "you@example.com",
9    date: datetime(year: 2025, month: 2, day: 1),
10 )
11
12 #show: style.with()
13
14 = Integration by Parts
15
16 #question(status: "done")[
17   Evaluate  $\int x e^x dx$ .
18 ]
19
20 #solution[
21   Using integration by parts with  $u = x$  and  $dv(x) = e^x dx$ :
22
23   
$$\int x e^x dx = x e^x - \int e^x dx = x e^x - e^x + C = e^x (x - 1) + C$$

24
25   
$$\int x e^x dx = x e^x - e^x + C$$

26
27   
$$\int x e^x dx = e^x (x - 1) + C$$

28
29 ]
```

7.2 Math-Heavy Solution with Custom Functions

```
1  #import "@local/gat-typst:0.1.0": *
2
3  #show: homework.with(
4    title: "Statistics Assignment",
5    course: "Probability Theory",
6    author: "Student",
7    email: "student@school.edu",
8  )
9
10 #show: style.with()
11
12 = Random Variables
13
```

```

14 #question(status: "done")[
15   Let  $X$  and  $Y$  be independent random variables.
16   Show that  $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ .
17 ]
18
19 #solution[
20   By definition:
21   $
22   
$$\text{Var}(X + Y) = \text{E}((X + Y)^2) - (\text{E}(X + Y))^2 \setminus$$

23   
$$= \text{E}(X^2 + 2XY + Y^2) - (\text{E}(X) + \text{E}(Y))^2 \setminus$$

24   
$$= \text{E}(X^2) + 2\text{E}(XY) + \text{E}(Y^2) - (\text{E}(X))^2 - 2\text{E}(X)\text{E}(Y) - (\text{E}(Y))^2$$

25   $
26
27   Since  $X$  and  $Y$  are independent,  $\text{E}(XY) = \text{E}(X) * \text{E}(Y)$ :
28   $
29   
$$\text{Var}(X + Y) = \text{E}(X^2) - (\text{E}(X))^2 + \text{E}(Y^2) - (\text{E}(Y))^2 \setminus$$

30   
$$= \text{Var}(X) + \text{Var}(Y)$$

31   $
32
33   #sub-solution[
34     *Note:* This result generalizes: for independent  $X_1, \dots, X_n$ ,
35     $
36     
$$\text{Var}(\sum_{i=1}^n X_i) = \sum_{i=1}^n \text{Var}(X_i)$$

37     $
38   ]
39 ]

```

7.3 Compact Cheatsheet

```

1  #import "@local/gat-typst:0.1.0": *
2
3  #show: cheatsheet.with(
4    title: "Midterm Cheatsheet",
5    course: "Linear Algebra",
6    author: "Your Name",
7    date: datetime(year: 2025, month: 3, day: 15),
8  )
9
10 #show: style.with()
11
12 #let gatbox = gatbox.with(color-cycle: true)
13
14 = Matrix Operations
15
16 #gatbox(title: [Transpose])[
17    $(A^T)_i = A_{(j \ i)}$ 
18
19    $(A \ B)^T = B^T \ A^T$ 
20 ]

```

```

21
22 #gatbox(title: [Determinant])[
23   $det(A B) = det(A) * det(B)$
24
25   $det(A^T) = det(A)$
26
27   $det(A^{(-1)}) = 1/det(A)$
28 ]
29
30 = Eigenvalues
31
32 #gatbox(title: [Definition])[
33   $A \text{ } vb(v) = \text{lambda } vb(v)$
34
35   $det(A - \text{lambda } I) = 0$
36 ]
37
38 #gatbox(title: [Properties])[
39   $product_{i=1}^n \text{lambda}_i = det(A)$
40
41   $sum_{i=1}^n \text{lambda}_i = "tr"(A)$
42 ]
43
44 = Vector Spaces
45
46 #gatbox(title: [Subspaces])[
47   *Null space:* $cal(N)(A) = Set(vb(x), A \text{ } vb(x) = vb(0))$
48
49   *Column space:* $cal(C)(A) = Set(A \text{ } vb(x), vb(x) \text{ in } \mathbb{R}^n)$
50
51   *Row space:* $cal(C)(A^T)$
52 ]
53
54 #gatbox(title: [Fundamental Theorem])[
55   $dim \text{ } cal(C)(A) + dim \text{ } cal(N)(A) = n$
56
57   $cal(C)(A)^{perp} = cal(N)(A^T)$
58 ]

```

8 Tips & Best Practices

Based on conventions and optimal usage of this package:

8.1 Math Notation

1. Use **semantic symbols** instead of generic ones:

```
1 // Good
2  $\textcolor{violet}{\$grad}$  f$ // Gradient
3  $\textcolor{violet}{\$product\_}(i=1)^n x\_i\$$  // Product
4
5 // Avoid
6  $\textcolor{violet}{\$nabla}$  f$
7  $\textcolor{violet}{\$Pi\_}(i=1)^n x\_i\$$ 
```

2. Space before subscripts with functions:

```
1 // Good
2  $\textcolor{violet}{\$V\_pi}$  (s)$ // Value function of policy  $\pi$  at state s
3  $\textcolor{violet}{\$Q\_theta}$  (s, a)$
4
5 // Bad (no space before parenthesis)
6  $\textcolor{violet}{\$V\_pi}$ (s)$
7  $\textcolor{violet}{\$Q\_theta}$ (s, a)$
```

3. Use **physica utilities** for common notation:

```
1  $\textcolor{blue}{\$vb}$ (x)$ // Vector bold
2  $\textcolor{blue}{\$vu}$ (n)$ // Unit vector
3  $\textcolor{blue}{\$cal}$ (A)$ // Mathcal letters (script)
4  $\textcolor{blue}{\$dd}$ (x)$ // Differential
5  $\textcolor{blue}{\$dv}$ (f, x)$ // Derivative
6  $\textcolor{blue}{\$pdv}$ (f, x, y)$ // Partial derivative
7  $\textcolor{blue}{\$iprod}$ (u, v)$ // Inner product (not angle brackets)
8  $\textcolor{blue}{\$norm}$ (x)$ // Norm (not  $\|x\|$ )
9  $\textcolor{blue}{\$abs}$ (x)$ // Absolute value
10  $\textcolor{blue}{\$eval}$ (f)_a^b$ // Evaluation bar
```

4. Leverage custom statistics functions:

```
1 // These are colored and use square brackets
2  $\textcolor{blue}{\$Ex}$ (X)$ // Expectation
3  $\textcolor{blue}{\$Var}$ (X)$ // Variance
4  $\textcolor{blue}{\$Pr}$ (A)$ // Probability
```

8.2 Layout

1. **Two-column homework** for dense problems:

```
1  $\textcolor{red}{\#show}$ : homework. $\textcolor{blue}{with}$ (
2   cols:  $\textcolor{red}{2}$ ,
```



```

3 // ... other params
4 )
5
6 // Override for specific solutions
7 #solution(cols: 1)[
8   This needs full width.
9 ]

```

2. **Color-cycle boxes** for organizing content:

```

1 #gatbox(color-cycle: true, title: "Theorem 1")[ ... ]
2 #gatbox(color-cycle: true, title: "Theorem 2")[ ... ]
3 #gatbox(color-cycle: true, title: "Theorem 3")[ ... ]
4 // Automatically cycles: blue → red → green → orange → purple

```

3. **Status tracking** for homework progress:

```

1 #question(status: "done")[Completed problem]
2 #question(status: "partial")[Work in progress]
3 #question(status: "todo")[Not started]

```

8.3 Organization

1. Use **sub-solutions** for complex multi-part problems:

```

1 #solution[
2   Main solution approach ...
3
4   #sub-solution[
5     For part (a), we have ...
6   ]
7
8   #sub-solution[
9     For part (b), note that ...
10  ]
11 ]

```

2. **Add notes** for clarifications or caveats:

```

1 #notes[
2   This assumes the matrix is invertible.
3   For singular matrices, use pseudoinverse.
4 ]

```

3. **Label equations** for cross-referencing:

```

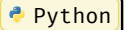
1 $
2   f(x) = x^2 + 2x + 1
3 $ <eq:quadratic>
4
5 From @eq:quadratic, we see that ...

```

8.4 Code

1. **Always specify language** for proper highlighting:

```
1 def example():  
2     pass
```

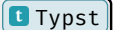


Not just:

```
1 def example():  
2     pass
```

2. **Use Callisto for notebook integration:**

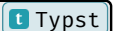
```
1 #let (render, Cell, In, Out) = callisto.config(  
2     nb: json("notebook.ipynb"),  
3 )  
4  
5 // Tag cells in notebook, then reference them  
6 #In("data-loading")  
7 #Out("plot-results")
```



8.5 Performance

1. **Compile time:** Heavy imports may slow compilation. Consider splitting large documents.
2. **Font Awesome:** Requires fonts installed system-wide. If icons don't render, install Font Awesome fonts.
3. **Cheatsheet margins:** 0.5cm is very tight. Adjust if printing issues occur:

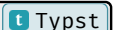
```
1 #set page(margin: 0.7cm) // Before cheatsheet.with()
```



8.6 Common Pitfalls

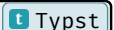
1. **Shorthand conflicts:** If you need literal * for multiplication:

```
1 $a ast b$ // Use 'ast' explicitly
```



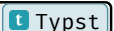
2. **Superscript confusion:** Remember ^* becomes star:

```
1 $x*$ // Renders as x* (star)  
2 $xast$ // Renders as x* (asterisk if needed)
```



3. **Column overrides:** Don't forget to override solution for full-width content in 2-column layouts:

```
1 #let solution = solution.with(cols: 1)
```



9 Reference

9.1 Quick Import Guide

```
1 // Full import
2 #import "@local/gat-typst:0.1.0": *
3
4 // Selective import
5 #import "@local/gat-typst:0.1.0": homework, question, solution, gatbox
6
7 // Import specific utilities
8 #import "@local/gat-typst:0.1.0": Ex, Var, KL, vb, cal
```

9.2 Template Comparison

Feature	style	homework	cheatsheet
Columns	N/A	1 or 2	3 (fixed)
Font size	N/A	13pt (1-col) / 10pt (2-col)	6pt
Headers	None	Optional	None
Margins	N/A	1 inch	0.5cm
Question boxes	No	Yes	No (manual)
Equation numbers	Yes (when used)	Yes	Yes

9.3 Function Reference

9.3.1 Styling

- `style(enable-shorthands: true, body)`

9.3.2 Templates

- `homework(title, course, university, author, email, ...)`
- `cheatsheet(title, course, university, author, email, ...)`

9.3.3 Components

- `question(status: "none", ...)`
- `solution(cols: 1, body)`
- `boxed-solution`
- `sub-solution(...)`
- `notes(...)`

9.3.4 Boxes

- `gatbox(color: black, color-cycle: false, ...)`
- `showybox(...)`

9.3.5 Math (Statistics)

- `Ex()` - Expectation
- `Var()` - Variance
- `Cov()` - Covariance

- `Corr()` - Correlation
- `Pr()` - Probability

9.3.6 Math (Other)

- `KL(p, q)` - KL divergence
- `refPol` - Reference policy
- `st, If, Else, Then` - Logic
- `clamp(x)`, `solve(x)` - Functions

9.3.7 Utilities

- `date-format(date)` - Format datetime
- `math-func(name, ...)` - Create custom math function
- `super-ast-as-star` - Show rule for *

9.4 Further Reading

For detailed package documentation:

- **Physica**: <https://github.com/Leedehai/typst-physics>
- **CeTZ**: <https://cetz-package.github.io/docs/>
- **Fletcher**: <https://github.com/typst/packages/blob/main/packages/preview/fletcher/>
- **Codly**: <https://typst.app/universe/package/codly>
- **Typst Documentation**: <https://typst.app/docs/>

9.5 Version History

v0.1.0 (Current)

- Initial release
- Templates: homework, cheatsheet
- Optional styling function with math shorthands, code highlighting, and table formatting
- Math utilities and custom box system
- Core package integrations (equate, physica, codly, booktabs, fontawesome)

For issues, suggestions, or contributions, see the project repository.